MIL-H-10120E
30 SEPTEMBER 1964
SUPERSEDING
MIL-H-10120D(MU)
21 MAY 1963

MILITARY SPECIFICATION

HEADHARNESS, C8R1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of protective mask headharness.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

JJ-W-155 — Webbing, Textile, (Cotton, Elastic).

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

D5-2-762 — Headharness, C8R1 — Assembly and Detail.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing D5-2-762 or on the subsidiary drawings shall conform to the specifications thereon

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or the specific characteristics set forth on the drawings, except as follows:

3.1.1 *Elastic webbing (JJ–W–155).*

3.1.1.1 *Elongation load.* The elongation load before and after aging shall be 36 ounces.

3.1.1.2 *Low-temperature elongation.* The webbing shall elongate not less than 50 percent after conditioning for 72 hours at −25° ± 2°F. and testing at −25° ± 2°F. with a load of 64 ounces, when tested as specified in 4.6.1.

3.1.1.3 *Permanent set at −25°F.* The webbing shall have a permanent set not to exceed 35 percent after being elongated 50 percent for 72 hours at −25° ± 2°F., when tested as specified in 4.6.2.

3.1.1.4 *Boiling water immersion.* The webbing shall meet the requirements of elongation before and after aging and at −25°F.; permanent set before and after aging and at −25°F.; and flexing; after boiling in water for not less than one hour, when tested as specified in 4.6.3.

3.1.1.5 *Flexing.* The webbing shall show no rupture of elastic threads nor more than 10 percent permanent set after flexing as specified in 4.6.4.

3.2 Manufacture and assembly. The headharness shall be manufactured and assembled to conform to Drawing D5–2–762.

3.2.1 *Pads.* The inner and outer pads shall be evenly cut and shall coincide without wrinkles or puckers when stitched together with the elastic webbing between them.

3.2.2 *Stitching.* When stitching is not continuous thread, it shall be back stitched no less than one-half inch. In sewing, the thread tension shall be properly maintained so that there will be no loose stitching, and the lock shall be embedded in the center of the sewn material.

3.2.3 *Clinch tips.* When attaching clinch tips, the elastic webbing shall be inserted the full depth of the tip. The pressure used in assembling the tips to the webbing shall not be sufficient to cut, break, or materially weaken the webbing. The tip shall form a smooth, flat, tight, and compact assembly at the end of the webbing.

3.3 Workmanship. The headharness shall be free from damage (tears, rips, holes, or deformed tips) and foreign matter (dirt, grease, or oil).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality of submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific

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inspection has not been provided in this specification have been satisfied.

4.2.3 *Inspection substation.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL–STD–105 are applicable to each substation and to the end item.

4.3 **Lotting.** A lot shall consist of the headharnesses offered for acceptance at one time, produced by one manufacturer, at one plant, under essentially the same manufacturing conditions. However, no more than one lot of elastic webbing shall be represented in any one lot of finished headharnesses.

4.4 **Sampling.**

4.4.1 *For examination.* Sampling shall be conducted in accordance with MIL–STD–105.

4.4.2 *For destructive tests.* Two pieces of webbing, no less than three yards long, shall be submitted from each lot of webbing for testing in accordance with 4.6.

4.5 **Inspection provisions.**

4.5.1 *For examination.* Sample headharnesses shall be examined in accordance with the classification of defects and with MIL–STD–105.

4.5.2 *For destructive testing.* Sample webbing shall be tested in accordance with 4.6. The acceptance number is zero.

4.5.3 *Classification of defects.*

4.5.3.1 *Headharness, C8R1 — assembly (Drawing D5–2–762).*

Categories	*Defects*
Critical:	None defined.
Major:	AQL 1.0 percent defective.
101	Component missing.
102	Sharp edges on metal components.
103	Component incorrectly assembled.
104	Workmanship (3.3).
105	Marking incorrect.
Minor:	AQL 2.5 percent defective.
201	Stitching incorrect.
202	Marking illegible.

4.6 **Tests.** Tests shall be conducted as follows:

4.6.1 *Low-temperature elongation.* Cut a 14-inch specimen from the webbing sample. Stretch to approximately 75 percent elongation three times slowly to break up the sizing. Mark the gage marks 10 inches apart in the approximate center of the specimen. Condition the specimen for the required period of time at the specified temperature. Following the conditioning period and maintaining the specified temperature, suspend the specimen vertically and attach the required tension load. Measure the distance between the gage marks within 15 seconds.

4.6.2 *Permanent set at −25°F.* Cut a 14-inch specimen from the webbing sample and mark off 10 inches, between gage marks, in the approximate center of the specimen. Stretch to break up sizing as in 4.6.1. Elongate the specimen as required and hold for the required period of time at the specified temperature; measure the extension and express as a percentage of the original distance.

4.6.3 *Boiling water immersion.* On an approximate 3-yard sample of the webbing, mark off seven 14-inch specimens. Place

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gage marks 10 inches apart on the webbing in the approximate center of each 14-inch specimen. Boil the sample for the required period of time in two liters of water contained in a 3-liter beaker, maintaining the level of water in the beaker. Remove the webbing and air dry at ambient temperature for a minimum of 24 hours. Cut the specimens apart and test as follows:

Requirement	*Test method*
Elongation:	
As received	**JJ–W–155**
After accelerated aging	*
At –25°F.	**4.6.1**
Permanent set:	
As received	**JJ–W–155**
After accelerated aging	*
At –25°F.	**4.6.2**
Flexing	**4.6.4**

*Substitute the following for the accelerated aging test method in JJ-W-155:

Accelerated aging (oxygen bomb). The test specimens selected for accelerated aging in the presence of oxygen shall be suspended for a continuous period of 96 hours in a closed chamber and subjected to oxygen having a pressure of 275 to 300 pounds per square inch and maintained at a temperature of 158° ± 2°F. At the end of the 96 hour period the test specimens shall be removed from the chamber and allowed to stand for 24 hours and then tested as specified in 4.6.1 and 4.6.2 of this specification.

4.6.4 *Flexing.* Hold the specimen horizontally on an apparatus equipped with a reciprocating arm arranged to transmit a direct horizontal elongation to the webbing and flex for 5,000 cycles, at a rate of 45 to 60 cycles per minute, under a 50-percent elongation. Examine the specimen visually for compliance with 3.1.1.5 and determine permanent set in accordance with applicable portion of JJ–W–155.

5. PREPARATION FOR DELIVERY

(For repair parts packaging see 6.5.)

5.1 Packaging, level C. Headharnesses shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Headharnesses packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. This headharness is intended for use as a component of the M9A1 and ABC–M17 masks.

6.2 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.3 Tests. It is suggested that destructive tests (see 4.6) on the webbing to determine compliance with 3.1.1.2, 3.1.1.3, 3.1.1.4 and 3.1.1.5 be conducted prior to manufacture of the headharness.

6.4 Toxicity. Any question raised regarding toxicity should be referred to the departmental medical authority. In the case of Army procurement, the Surgeon General will act as advisor to the procuring agency.

6.5 Repair parts packaging. When repair parts packaging and packing is required for items authorized for stocking as repair parts, the repair parts packaging data sheet covering the C8R1 Headharness should be used.

(Copies of repair parts packaging data sheets may be obtained from the Director for Engineering

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and Industrial Services, ATTN: Chief, Design Branch, U.S. Army Edgewood Arsenal, Md. 21010)

6.6 International interest. Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Army—MU

Navy—SH

Air Force—84

International interest (see 6.6)

Review activities:

Army—MD

Navy—SH

User activities:

Navy—MC

Air Force—84

Preparing activity:

Army—MU(EA)

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